Proof. Let $C_1, C_2, \ldots, C_n$ be the conjugacy classes of G. Let g_{C_i} be representatives of C_i. Set

$$\lambda_i = \chi_V(g_{C_i}) \frac{|C_i|}{\dim V}.$$

Proposition 5.3.2. *The numbers λ_i are algebraic integers for all i.*

Proof. Let C be a conjugacy class in G, and let $P = \sum_{h \in C} h$. Then P is a central element of $\mathbb{Z}[G]$, so it acts on V by some scalar λ, which is an algebraic integer (indeed, since $\mathbb{Z}[G]$ is a finitely generated $\mathbb{Z}$-module, any element of $\mathbb{Z}[G]$ is integral over $\mathbb{Z}$, i.e., satisfies a monic polynomial equation with integer coefficients). On the other hand, taking the trace of P in V, we get $|C|\chi_V(g) = \lambda \dim V$, $g \in C$, so $\lambda = \frac{|C|\chi_V(g)}{\dim V}$. $\square$

Now, consider

$$\sum_i \lambda_i \overline{\chi_V(g_{C_i})}.$$

This is an algebraic integer, since:

(i) λ_i are algebraic integers by Proposition 5.3.2,

(ii) $\chi_V(g_{C_i})$ is a sum of roots of unity (it is the sum of eigenvalues of the matrix of $\rho(g_{C_i})$, and since $g_{C_i}^{|G|} = e$ in G, the eigenvalues of $\rho(g_{C_i})$ are roots of unity), and

(iii) $\overline{\mathbb{Z}}$ is a ring (Proposition 5.2.4).

On the other hand, from the definition of λ_i,

$$\sum_{C_i} \lambda_i \overline{\chi_V(g_{C_i})} = \sum_i \frac{|C_i|\chi_V(g_{C_i})\overline{\chi_V(g_{C_i})}}{\dim V}.$$

Recalling that χ_V is a class function, this is equal to

$$\sum_{g \in G} \frac{\chi_V(g)\overline{\chi_V(g)}}{\dim V} = \frac{|G|(\chi_V, \chi_V)}{\dim V}.$$

Since V is an irreducible representation, $(\chi_V, \chi_V) = 1$, so

$$\sum_{C_i} \lambda_i \overline{\chi_V(g_{C_i})} = \frac{|G|}{\dim V}.$$